

Architectural Design Document: Network-Aware Mine Haulage Prediction

Project: GNN-Based Mine Haulage Prediction with Failure Awareness and Uncertainty

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1 Executive Summary & Problem Formulation

In open-pit mine haulage operations, predicting cycle times accurately across a road network is critical for automated truck dispatching and short-term scheduling. Traditional mining operations rely on flat, tabular machine learning models (e.g., XGBoost) or static empirical physics models. However, haulage networks operate as dynamic, directed graphs where equipment failures, queue buildups, and localized congestion propagate spatially across road segments. This design document establishes the empirical limitations of tabular modeling based on our Week 1 benchmark findings on 4,800 historical haulage observations and justifies the progressive transition from a **Graph Convolutional Network (GCN)** to a **Graph Attention Network (GAT)** integrated with Deep Ensemble uncertainty quantification.

2 Empirical Justification & Tabular Baseline (Week 1)

Our Week 1 benchmark evaluation utilized an XGBoost regressor on 4,800 historical haulage observations. **Methodology & Rigor:** Evaluations were conducted using a strict 80/20 train-test split (random seed = 42). To guarantee zero data leakage, observations were strictly isolated by operating episodes (shifts), ensuring no temporal or spatial cross-contamination of arc states between the training and testing sets. Performance metrics (evaluated in seconds) are summarized below:

Trip Type	MAE (s)	RMSE (s)	Syllabus Target (MAE)
Loaded Trips	85.76	110.88	< 80.00
Empty Trips	71.37	91.98	N/A
Combined Trips	78.72	101.72	< 110.00

While the baseline successfully captures deterministic mechanics, an analysis of the model’s feature importance hierarchy revealed a critical empirical limitation:

Feature Category	Primary Variables	Importance	Operational Interpretation
Deterministic Physics	loaded, max_gradient_pct	65.89%	Tabular trees excel at static vehicle mechanics (engine power vs. gravity).
Global State Flag	failure_active	25.16%	Detects <i>when</i> a failure occurs, but applies a spatially invariant penalty.
Network Congestion	congestion_level, payload_t	3.36%	The Tabular Blind Spot: Fails to utilize localized queue states.

The Architectural Flaw: Why did `congestion_level` account for just 3.36% of model importance when physical traffic is the primary operational cause of mine haulage delays? Because a tabular tree treats observations as independent, identically distributed (*i.i.d.*) rows. When an equipment failure (`failure_active = 1`) occurs at Crusher CR, XGBoost cannot trace the directed road segments to determine *which* approaching trucks are trapped in the resulting cascade vs. which trucks are operating on uncongested waste dumps across the pit. To close the remaining **5.76 s error gap** and hit our **<80.00 s Loaded MAE KPI target**, we must transition to an architecture natively capable of spatial message passing across a directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$.

3 GCN Baseline & Edge-Level Prediction Architecture

For our Week 2 baseline, we implement a 2-layer Graph Convolutional Network (Kipf & Welling, 2017). A standard GCN updates node feature representations $\mathbf{H}$ using the isotropic propagation rule:

$$\mathbf{H}^{(l+1)} = \sigma \left(\tilde{\mathbf{D}}^{-\frac{1}{2}} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{-\frac{1}{2}} \mathbf{H}^{(l)} \mathbf{W}^{(l)} \right) \quad (1)$$

where $\tilde{\mathbf{A}}$ is the adjacency matrix with added self-loops and $\tilde{\mathbf{D}}$ is the diagonal degree matrix.

The Limitation for Mining Operations: Standard GCNs utilize **symmetric, isotropic aggregation** ($\tilde{\mathbf{D}}^{-\frac{1}{2}} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{-\frac{1}{2}}$). This formulation forces a node to aggregate information from all adjacent neighbors with static weights dictated purely by topological degree. In a mine haulage network, an intersection connected to two empty waste dumps and one bottlenecked ore shovel will weigh messages from all three destinations roughly equally. Standard GCNs lack the capacity to dynamically prioritize active operational bottlenecks.

Edge-Level Prediction Head

Standard GCN and GAT layers natively produce *node-level* embeddings ($\mathbf{h}_i$), yet our objective is to predict travel time across a directed route. To bridge this gap, we will implement an edge-level prediction head. For any directed haulage movement from source node u to destination node v , we extract their respective final-layer embeddings $\mathbf{h}_u$ and $\mathbf{h}_v$. These are concatenated with the physical directed edge features $\mathbf{e}_{uv}$ (e.g., total distance, maximum gradient) and passed through a Multi-Layer Perceptron (MLP) to compute the final scalar travel-time prediction $\hat{y}_{uv}$:

$$\hat{y}_{uv} = \text{MLP}(\mathbf{h}_u \parallel \mathbf{h}_v \parallel \mathbf{e}_{uv}) \quad (2)$$

This formulation ensures that the deterministic physics of the road segment interact directly with the dynamic, network-aware congestion states learned by the node embeddings.

4 Proposed Architecture Hypothesis: Graph Attention Networks (GAT)

To overcome the isotropic aggregation of the GCN baseline, we propose evaluating a **Graph Attention Network (GAT)** (Velićković et al., 2018). GATs replace fixed degree-normalization with learnable, anisotropic attention coefficients α_{ij} , which compute the relative importance of neighbor j 's features onto node i :

$$\alpha_{ij} = \frac{\exp(\text{LeakyReLU}(\mathbf{a}^T[\mathbf{W}\mathbf{h}_i \parallel \mathbf{W}\mathbf{h}_j]))}{\sum_{k \in \mathcal{N}_i} \exp(\text{LeakyReLU}(\mathbf{a}^T[\mathbf{W}\mathbf{h}_i \parallel \mathbf{W}\mathbf{h}_k]))} \quad (3)$$

Research Hypothesis: Expected Advantages over GCN

We hypothesize that the GAT architecture will outperform the GCN baseline in failure-aware prediction scenarios due to the following expected advantages:

1. **Dynamic Bottleneck Routing:** We expect the attention mechanism to dynamically learn to heavily attend ($\alpha_{ij} \rightarrow 1$) to nodes exhibiting high queue capacities, isolating congestion impacts more effectively than a symmetric GCN.
2. **Failure-Aware Propagation (Week 4 Integration):** During simulated equipment failures, GAT layers should mathematically shift attention weights across the graph, capturing traffic rerouting impacts without the rigid topological constraints of a standard GCN.
3. **Operational Interpretability (Week 3 Deliverable):** By extracting the top-3 attention weights per arc, we aim to map attention heatmaps onto the spatial mine layout, testing if the model's learned routing logically aligns with physical dispatch bottlenecks.

5 Epistemic Uncertainty & Downstream Optimizer Integration

To support the downstream Two-Stage Stochastic MILP truck-dispatch optimizer (Week 7), point-predictions of travel time are insufficient. The optimizer requires calibrated probability distributions to manage risk.

We will implement a **Deep Ensemble of $M = 10$ independently seeded GAT networks**. For any directed haulage arc, the ensemble mean μ serves as the expected cycle time, while the sample variance σ^2 captures **epistemic uncertainty** (model uncertainty due to limited training data in specific failure states). Note that capturing *aleatoric uncertainty* (inherent data noise) falls outside the scope of this ensemble and would require a dedicated probabilistic loss formulation.

$$\mu = \frac{1}{M} \sum_{m=1}^M \hat{y}_m, \quad \sigma^2 = \frac{1}{M-1} \sum_{m=1}^M (\hat{y}_m - \mu)^2 \quad (4)$$

Optimizer Integration: This quantified epistemic uncertainty (σ^2) directly informs the formulation of discrete **Sample Average Approximation (SAA) scenarios** within the MILP optimizer. Rather than relying on automatic or arbitrary adjustments, extreme travel-time realizations sampled from high-variance arcs will act as explicit constraints or risk terms in the stochastic objective function, penalizing the optimizer for routing trucks through mathematically uncertain corridors during failure events.